Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 3F nr. 1.12

$$f(x) = \frac{x^3 + 4x^2 + x + 1}{x - 3}$$

VA:

$$\lim_{x \to 3+} \frac{x^3 + 4x^2 + x + 1}{x - 3} = +\infty$$
$$\lim_{x \to 3-} \frac{x^3 + 4x^2 + x + 1}{x - 3} = -\infty$$

VA: $x = 3$

x		3	
$f(x)$	$+$	$\vert$	$-$

HA:

$$\lim_{x \to \infty} \frac{x^3 + 4x^2 + x + 1}{x - 3} = \lim_{x \to \infty} \frac{x^3}{x} = \lim_{x \to \infty} x^2 = +\infty \Rightarrow \text{geen HA}$$

SA:

$$\lim_{x \to \infty} \frac{\frac{x^3 + 4x^2 + x + 1}{x - 3}}{x} = \lim_{x \to \infty} \frac{x^3 + 4x^2 + x + 1}{x(x - 3)}$$
$$= \lim_{x \to \infty} \frac{x^3 + 4x^2 + x + 1}{x^2 - 3x} = \lim_{x \to \infty} \frac{x^3}{x^2} = \lim_{x \to \infty} x = \infty \Rightarrow \text{geen SA}$$

References